# Alex Spaulding

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## Professional Summary

Computer Science student building systems software, high-performance applications, and AI-integrated tools. Strong foundation in C, Rust, and TypeScript with a focus on systems design, low-level optimization, and AI integration. Seeking Software Engineering internship opportunities.

## Education

**Eastern Washington University** *Spokane, WA*

*Bachelor of Science (BS) in Computer Science, Math Minor* · Feb 2025 - Present

- **Relevant Coursework**: Algorithms, Computer Architecture, C & UNIX Programming, Cybersecurity, Relational Databases, Computer Networks

**The University of Montana** *Missoula, MT*

*Some College* · Sep 2021 - 2024

- **Relevant Coursework**: Data Structures, Software Principles I & II, Operating Systems & Command Line Interface

## Technical Skills

**Languages**: C, C++, Rust, Swift, Python, Java, JavaScript/TypeScript, SQL, Bash, Nix
**Systems & Security**: Unix, Inter-Process Communication (IPC), macOS/iOS internals, Reverse Engineering, Mach-O, Kernel/Runtime Mods, Git, Continuous Integration/Deployment (CI/CD)
**AI Integration**: OpenAI APIs, Large Language Model (LLM) integration, Whisper, GPT, Text-to-Speech synthesis
**Frontend & Backend**: React, Next.js, Node.js, REST APIs, Docker, Amazon Web Services (AWS), Google Cloud Platform (GCP), HTML/CSS

## Projects

**apple-sharpener (269 ★)** *GitHub*

*macOS System Customization (Objective-C, C)* · 2023 - Present

- Built a macOS customization tool (**269 stars**); resolved **15+ critical bugs** via low-level profiling, Objective-C runtime hooks, and Mach-O patching.

**Wawona (176 ★)** *Wawona.io* / *GitHub*

*World's First iOS Wayland Compositor (Rust, C, Swift)* · 2025 - Present

- Founded Wawona and developed the **world's first iOS Wayland compositor**, achieving full Apple/Android platform parity; managed **176+ stars** and **22+ active issues/PRs**.
- Engineered Wawona.io using **Zola/GSAP** and automated testing via **TestFlight/Fastlane** for deployments.

**Whisperer** *GitHub*

*AI Voice Assistant for watchOS (Swift, Node.js)* · 2023 - Present

- Developed an AI watchOS client with OpenAI Whisper and **markdown rendering**; utilized my **Base16/24** library TintedThemingSwift.

## Experience

**University of Montana IT Help Desk** *Missoula, MT*

*IT Consultant* · Sep 2020 - Jan 2025

- Resolved **1,200+ service tickets** and supported **20+ walk-ins weekly**; utilized **Jamf/DevOps tools**, authored **50+ technical guides**, and managed Windows activations/purchase requests.
- Maintained campus hardware including **50+ projector and thin client troubleshooting** cases, hard drive wipes, and e-waste disposal.

**Sunburst Sensors** *Missoula, MT*

*Assembly Technician* · Jan 2022 - Mar 2025

- Assembled pCO2/pH instruments via PCB soldering, **precision fiber optic cleaving**, epoxy potting, and machining; fabricated brass and optical filters and spot welded **300+ battery packs**.
- Executed bulk firmware updates for **1,000+ microcontrollers**; utilized proprietary software for sensor calibration and executed hydrostatic pressure testing to verify integrity.

**Wheelbound Productions** *Missoula, MT*

*Web Developer* · Apr 2018 - Sep 2018

- Launched **700+ responsive web pages** for dumonds.com implementing mobile-optimized design patterns; enhanced performance for high-traffic production sites.

## Leadership & Awards

**VP – Association for Computing Machinery (ACM)** – Led technical events, Hackathons, and Capture The Flag (CTF) competitions for the student community · 2025 - Present

**1st Place – Gonzaga University Hackathon** – DevOps Lead; architected CI/CD pipelines for real-time game engine · 2025

**1st Place – UM Cyber Competition** – CTF winner; performed network reconnaissance and exploitation · 2024

**Front End Certificate** – 18-week program; 18 projects including a (Final Project) · Jul 2021

*References available upon request.*